

PROJECT CHARTER

Project Title	PPT Outline Intelligent Generation and Content Completion System		
Date of Authorization	March 30, 2026		
Project Start Date	March 30	Project Finish Date	June 14
Key Schedule Milestones: Milestone 1: System Architecture and Solution Design Completed. Finalize the LLM prompt strategies and the RAG knowledge base retrieval architecture. Milestone 2: Core Generation Module Developed. The system successfully generates a structured presentation outline based on the user's input topic. Milestone 3: RAG and Content Completion Integrated. External information retrieval is operational, and logically clear content is populated into individual slide pages. Milestone 4: End-to-End Workflow Operational. The complete workflow, from user input to final result display, is successfully executed. Milestone 5: System Evaluation and Acceptance Passed. Testing with 10 users confirms the generation time is reduced to 20 minutes or less. Factual accuracy reaches 90% or higher , and user satisfaction scores are at least 4.0 out of 5			
Budget Information: This project is primarily course-based and does not involve a dedicated financial budget. The main resources required are existing development tools, open-source LLM APIs, and team members' labor. Estimated external costs are minimal, limited to potential API usage fees, which are expected to be covered by team members individually. All software and libraries used are open source or available through the university.			
Project Manager: Yiwei Chen, 2350217@tongji.edu.cn, 18221401689			
Project Objectives: 1. To design and develop a prototype system that can generate a structured PPT outline based on a user's topic or input text. 2. To enable the system to organize presentation content into clear sections and a logical slide structure. 3. To incorporate a RAG-based retrieval module so that external knowledge can be used to support content generation. 4. To improve the richness and factual support of slide content compared with a text-only generation approach. 5. To provide a basic end-to-end process including user input, outline generation, content completion, and result presentation. 6. To verify the feasibility and practical value of the prototype in typical academic and office presentation scenarios.			

Main Project Success Criteria:

1. A working prototype is completed by 14 June 2026 within the planned project schedule.
2. The system is able to accept at least two types of input, including a short topic and a longer text or document.
3. For a valid input, the system can generate a PPT outline with at least 5 slides and a clear logical structure.
4. The retrieval module is successfully integrated, and in at least 80% of the selected test cases, the generated content includes relevant supporting information from external sources.
5. In comparative testing, the RAG-based version performs better than the baseline LLM-only version in terms of content relevance, informativeness, and factual support in most test cases.
6. All required course deliverables, including the prototype, RBS, WBS, total work-hour estimation, and planning tool, are completed and submitted on time.

Approach:

This project will be carried out in an iterative and prototype-based manner. At the beginning of the project, the team will further clarify the project scope, user needs, and major requirements according to the identified problem and project goal. Based on this, the team will complete requirement analysis and task decomposition through the RBS and WBS, and prepare an overall project plan according to the course schedule.

During the development stage, the project will be implemented step by step. The main work includes input analysis, PPT outline generation, retrieval module integration, content completion, and basic result presentation. In order to better show the value of retrieval enhancement, the team will first build a baseline version based on LLM-only generation, and then develop a RAG-based version for comparison.

In terms of project organization, team members will divide the work according to different responsibilities, such as requirement analysis, system design, prototype implementation, testing, and documentation. Regular meetings will be held to check project progress, discuss current problems, and make necessary adjustments to the plan. This will help the team control the schedule and reduce the risk of delay.

For quality assurance, the team will conduct internal reviews and functional testing throughout the development process. After the prototype is completed, representative test cases from academic and office presentation scenarios will be selected for evaluation. The evaluation will mainly focus on outline structure, logical coherence, content relevance, and factual support. The final results will be summarized in the project report and presentation.

Requirements Breakdown Structure (RBS):

The requirements for the prototype system are categorized into three primary functional domains to ensure a logical flow from user input to the final presentation output:

1. Outline Generation Requirements

1.1 User Intent Parsing Feature: The system must accurately analyze the user's provided topic or uploaded source documents to identify key themes.

1.2 Structured Outline Generation Feature: The system must generate a hierarchical outline with a clear logical flow, ensuring at least 5 slides are produced for any valid input.

1.3 Multi-Format Input Support: The system must support at least two types of input: short text strings (topics) and longer text files or documents.

2. Content Expansion & Enrichment Requirements

2.1 RAG Document Retrieval and Vectorization Feature: The system must be able to index external knowledge and retrieve relevant information using a Retrieval-Augmented Generation (RAG) architecture.

2.2 Fact-Checking and Content Integration Feature: The system must cross-reference generated content with retrieved sources to ensure a factual accuracy rate of 90%.

2.3 Contextual Informativeness: The system must improve the richness of the slides compared to a standard LLM-only approach by incorporating specific supporting details from external data.

3. Final Output & User Interaction Requirements

3.1 Slide Layout and Content Assembly Feature: The system must organize the final enriched content into a slide-ready format, including sections and bullet points.

3.2 End-to-End Workflow Execution: The system must provide a seamless process from the initial user prompt to the final result display.

3.3 Performance and Latency Management: The system must generate the final output quickly enough to reduce total slide creation time to ≤ 20 minutes for a 10-slide deck.

Work Breakdown Structure (WBS):

Total Estimated Effort: Approximately 150 hours

1. Project Definition and Planning (10 hours)

- Draft and finalize the Project Charter (4 hours)
- Refine RBS and WBS structures (6 hours)

2. System Design Phase (25 hours)

- Design RAG knowledge base architecture and data processing pipeline (10 hours)
- Design LLM prompt templates and outline generation logic (10 hours)
- Evaluate and select the underlying PPT generation library (5 hours)

3. System Development and Implementation Phase (75 hours)

- Develop core RAG retrieval and vector database integration modules (20 hours)
- Develop LLM structured outline and slide content generation modules (25 hours)
- Develop PPT file assembly and formatting modules (15 hours)
- Integrate basic user interaction workflow for UI or CLI (15 hours)

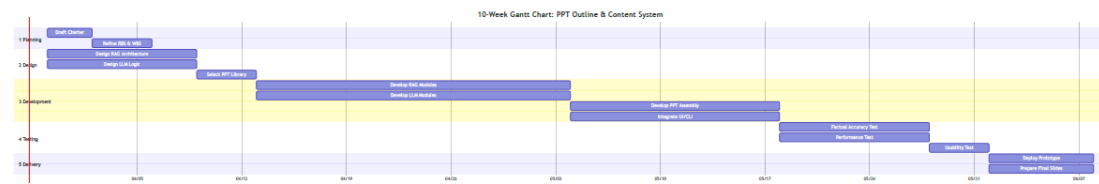
4. Testing and Evaluation Phase (30 hours)

- Conduct factual accuracy testing to verify against RAG sources (12 hours)
- Conduct generation time performance testing (8 hours)
- Conduct user usability testing and post-use questionnaires (10 hours)

5. Delivery and Presentation (10 hours)

- Deploy the prototype system and prepare the demonstration (5 hours)
- Prepare the project summary and final presentation slides (5 hours)

Gantt Chart:



Roles and Responsibilities

Name	Role	Position	Contact Information
Yiwei Chen	Project Manager	Manager, Programmer	2350217@tongji.edu.cn
Yicheng Huang	Team Member	Programmer	huangyicheng27@tongji.edu.cn
Guang Yang	Team Member	Programmer	2251756@tongji.edu.cn
Sayu Chen	Team Member	Programmer	2354447@tongji.edu.cn

Sign-off:

Yiwei Chen Yicheng Huang Guang Yang Sayu Chen